

Properties of Eigenvalues and Eigenvectors

Algebraic Multiplicity

Defn. The **algebraic multiplicity** of eigenvalue λ is its multiplicity as a root of the characteristic polynomial.

Fact. The dimension of the eigenspace of λ is at most its algebraic multiplicity.

Eigenvectors and Linear Independence

Fact. *Eigenvectors for distinct eigenvalues are linearly independent.*

Eigenvalues of Matrix Powers

Fact. *If matrix A has eigenvalues λ_i , then the power A^k has eigenvalues λ_i^k . Moreover, the eigenvectors are the same.*

Trace and Determinant

The **trace** of a matrix is defined as the sum of the diagonal entries.

Fact. For any matrix A ,

(a) the determinant of A equals the product of its eigenvalues.

(b) the trace of A equals the sum of its eigenvalues.

Complex Conjugates

Recall that i denotes the square-root of -1 .

Defn. If $\lambda = a + bi$, then its **(complex) conjugate** is $a - bi$.

Complex Eigenvalues

Fact. *If λ is a complex eigenvalue of A , then so is its conjugate.*

An Example

Consider the matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.

The characteristic polynomial is $(a - \lambda)^2 + b^2$; eigenvalues are $\lambda = a \pm bi$.

As a matrix transform, this represents scaling by $\sqrt{a^2 + b^2}$ and rotation through $\arctan b/a$.

Symmetric Matrices

Fact. *A real symmetric matrix has only real eigenvalues.*

Summary

The algebraic multiplicity of eigenvalue is its multiplicity as root of the characteristic polynomial; the eigenspace has dimension at most its algebraic multiplicity.

Eigenvectors for distinct eigenvalues are linearly independent.

If A has eigenvalue λ_i , then A^k has eigenvalue λ_i^k with same eigenvector.

Summary (cont)

The product of the eigenvalues is the determinant; the sum of the eigenvalues is the trace, which is the sum of the diagonal entries.

If λ is complex eigenvalue of real matrix, then so is its conjugate. A real symmetric matrix has real eigenvalues.